Physical Chemistry (I)

# Lecture 16

2차 중간고사 안내 및 복습

# 2차 중간고사 안내

- 시간: 5월 13일(화) 오전 10:30~11:30
- 장소: 화학관 112호(홀수 학번), 315호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

# 시험 범위

- 5장 “헬름홀츠 및 기브스 에너지”
- 6장 “통계역학”
- 7장 “상 전이”
- 8장 “단순 혼합물” 중 진도 나간 곳까지

# 시험 문제

총 여섯 문제

- 과제 문제 혹은 그 변형(3문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(3문제)
  - 강의 자료 기준

## Chapter 5

### Thermodynamic potentials

**Internal energy**
$U$

**Helmholtz energy**
$A = U - TS$

**Enthalpy**
$H = U + pV$

**Gibbs energy**
$G = H - TS$

Clausius ineq. → **Spontaneity**

$\Delta A_{T,V} \leq 0$

$\Delta G_{T,p} \leq 0$

First law → **Maximum work**

$A_T = w_{\text{max}}$

$G_{T,p} = w_{\text{add,max}}$

**Differential forms**

**Applications**

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

**Thermochemistry**

Pressure dependence

# Concepts and Derivations in Chpt 5

- [C] Helmholtz and Gibbs energies
- [D] Spontaneity: Clausius inequality
- [D] Meaning of Helmholtz and Gibbs energies
- [D] Thermodynamic potentials
- [D] Gibbs-Helmholtz equation
- [D] Maxwell relations
- [D] Thermodynamic equation of state
- [D] Pressure dependence of Gibbs energy

## Chapter 6

### Classical Stat Mech

$$S = k_\mathrm{B} \ln W$$

**Boltzmann factor**

$$P(s) = \frac{1}{z} e^{-E(s)/k_\mathrm{B}T}$$

- Partition function

$$z = \sum e^{-E(s)/k_\mathrm{B}T}$$

**Phase space**

**Partition function**

$$z = \frac{1}{h^3} \iint e^{-E/k_\mathrm{B}T}\, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

- Many-particle PF

$$Z = \frac{1}{N!} z^N$$

**Perfect gas**

- Partition function
- Helmholtz energy
- Pressure

Perfect gas law

**Real gas**

**Equipartition theorem**

- Partition function
- Internal energy

**Thermodynamics**

Helmholtz energy, internal energy, …

# Concepts and Derivations in Chpt 6

- [D] Boltzmann factor
- [C] Partition function
- [D] Partition function and thermodynamics
  - [D] Internal energy
  - [D] Shannon entropy
  - [D] Helmholtz energy
  - [D] Other thermodynamic quantities
  - [D] Example: spin system
- [C] Phase space

# Concepts and Derivations in Chpt 6

- [C] Partition function of classical system
  - [D] 1-dimensional spring
  - [D] $N$-particle partition function
  - [C] Indistinguishability
- [D] Perfect gas: perfect gas law
- [C] Real gas: van der Waals equation
- [D] Equipartition theorem

## Chapter 7

### Concepts

- Phases
- Phase transitions
- Phase diagrams

### Phase stability

- Equilibrium

$$\mu_1 = \mu_2$$

- Spontaneous

$$\mu_1 > \mu_2$$

### Phase rule

$$F = C - P + 2$$

### van der Waals fluid

#### Liquid-gas transition

- Phase coexistence
- Metastable phases

#### Supercritical fluid

#### Critical point

- Critical constants
- Principle of corresponding states

### Chemical potential

$$\mu = G/n$$

Modification of thermodynamic potentials

#### Gibbs-Duhem equation

$$\mathrm{d}\mu = -S_\mathrm{m}\mathrm{d}T + V_\mathrm{m}\mathrm{d}p$$

### Generalization

- Phase transitions
- Phase boundaries

# Concepts and Derivations in Chpt 7

- [C] Phases and phase diagrams
- [C] Phase transitions
- [C, D] Chemical potential
- [D] Rewriting thermodynamic potentials
- [D] Gibbs-Duhem equation
- [C, D] Phase rule
- [C] Liquid-gas transitions

# Concepts and Derivations in Chpt 7

- [C] vdW: Liquid-gas transitions
  - [C] Maxwell construction
  - [D] Lever rule
  - [C] Binodal and spinodal lines
- [C] vdW: Supercritical fluid
- [C] vdW: Critical point
  - [D] Critical constants
  - [D] Critical compression factor
  - [D] Reduced variables
  - [C] Principle of corresponding states

# Concepts and Derivations in Chpt 7

- [D] $T$- and $p$-dependences of chemical potentials
- [C] Phase boundaries
  - [D] Clausius equation
  - [D] Clausius-Clapeyron equation
  - [C] Interpretations

## Chapter 8

- Phase
- Chemical potential
- Gibbs-Duhem eq.

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**Multi-component system**

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**Gas mixture**

- Dalton’s law
- Perfect gas
- Real gas

# Concepts and Derivations in Chpt 8

- Extension to multi-component systems
  - [C] Thermodynamic potentials
  - [C, D] Partial molar quantities
  - [C, D] Chemical potential
  - [D] Gibbs-Duhem equation
- [C] Dalton's law
- [D] Mixing of perfect gases
- [C] Real gases: fugacity and fugacity coefficient
- [D] Mixing of real gases